GlandersWatch: A Modular Platform for Glanders Risk Prediction, Retrieval-Augmented Explanation and Owner Alerting

Project Report Submitted

in partial fulfillment of the requirements for the award of the

Bachelor of Technology

in

Computer Science Engineering

by

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November 2026

Certificate

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Dr. Sangita Dhara
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Acknowledgments

We would like to express our sincere gratitude to our respected supervisor, Dr. Sangita Dhara, for the invaluable guidance, support and encouragement given throughout this project. Her insights and expertise were instrumental in shaping this work and helped us overcome numerous challenges. We are also thankful to the faculty and staff of the Department of Computer Science and Engineering, International Institute of Information Technology, Bhubaneswar, for providing the resources and a conducive learning environment. Finally, we would like to thank our families and friends for their unwavering support and understanding during the course of this project.

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Abstract

Glanders is a bacterial zoonosis of equids caused by *Burkholderia mallei*. It is notifiable, frequently fatal, and remains endemic in parts of Africa, Asia and South America. Indian surveillance between 2019 and 2023 identified 728 seropositive equids across 138 districts, with strong geographic concentration. The published machine learning literature on this disease is confined to laboratory identification of the organism by spectroscopic methods, and no population-level geographic risk model has been reported.

This report presents GlandersWatch, a platform built from three independently deployable components: a calibrated machine learning risk engine, a retrieval-augmented layer that explains each prediction with citations to official guidance, and an alert service that delivers the result to equine owners. The decomposition is the contribution as much as the model is. Each component communicates through a versioned schema rather than shared internals, none hardcodes the disease, and each degrades without failing the others.

The risk engine combines four public datasets not previously applied to glanders into a country-year panel: a text-mined occurrence corpus from Europe PMC, FAOSTAT livestock stocks, World Bank development indicators and NASA POWER reanalysis climatology. The panel contains 3,078 country-year observations across 162 equine-holding countries from 2005 to 2023, with 172 positive labels and 44 engineered features. Six model families are trained and compared under three validation schemes.

On a forward temporal holdout the strongest model reaches a precision-recall AUC of 0.554 against a base rate of 0.056, roughly a ten-fold lift, with an expected calibration error of 0.016. Performance falls to 0.304 under leave-country-out validation, which is the estimate to quote for an unseen country. The model performs least well in low-income regions, where label sparsity limits what can be learned. The dominant limitation is that the label reflects documented activity rather than true incidence, so the model largely learns where disease is reported. Reporting effort is the principal confounder and the main target for future work. The explanation and alert components are specified to the level of their interfaces, rule semantics and evaluation protocol, and are not reported as measured results because they are not deployed.

Keywords: Glanders, *Burkholderia mallei*, Risk Prediction, Gradient Boosting, Probability Calibration, SHAP, Retrieval-Augmented Generation, Citation Grounding, Alert Fatigue, Modular System Design, MLOps, Animal Health Surveillance.

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Chapter 1

Introduction

1. Background

Glanders is a contagious and often fatal disease of horses, donkeys and mules caused by *Burkholderia mallei*. The organism has no environmental reservoir, and asymptomatic carriers remain infectious for long periods. Human infection is rare but carries a high case fatality rate and follows occupational contact. The disease is notifiable to the World Organisation for Animal Health and is classified as a potential bioterrorism agent.

The economic weight sits in low and middle income countries, where equids remain essential to transport, construction and agriculture. India recorded 5.44 lakh equids in the 2019 Livestock Census, many held by households that depend on them for daily income, so losing a working animal is a direct loss of livelihood. Early warning is useful even where mass treatment is not feasible.

Surveillance shows the disease is neither rare nor geographically contained. A national programme screening 158,798 sera from 391 districts in 2019 to 2023 found 728 seropositive animals in 138 districts, roughly sixty per cent of them in Uttar Pradesh; earlier surveillance covering 2006 to 2018 reported 1,145 positives among 128,404 sera.

2. Problem Definition

A district veterinary officer faces a narrow question: is glanders likely to appear in this area in the coming year? Answering it means combining the history of the disease nearby, the size of the susceptible population, and the conditions that permit transmission.

Surveillance reports what has already happened, and forecasting systems exist for several livestock diseases but not for glanders. A prediction alone would still not be usable: an owner cannot act on a probability of 0.24, and an officer cannot defend a forecast whose basis is hidden. A warning that arrives for every area every month also teaches its recipients to ignore it.

The problem therefore has four parts. Two concern producing the prediction, two concern turning it into something a person can act on.

1. **Data and label.** No public source records glanders occurrence completely, so occurrence must be reconstructed from the literature and joined to population and environmental covariates at a common resolution. The outcome is then rare, so accuracy is useless and a negative observation needs an explicit meaning.
2. **Model selection under distribution shift.** A model fitted on some countries must be evaluated on countries it has never seen, because that is the deployment condition.
3. **Grounded explanation.** The prediction must be explained in words a non-specialist can act on, every claim must trace to official control guidance, and the layer must refuse rather than generate an answer the corpus does not support.

4. **Delivery without alert fatigue.** A probability is not an alert. The operating point and the message rules must keep volume low enough that messages are read, while the threshold stays defensible against the events caught.

3. Objectives

There is one research objective and three engineering objectives.

1. **Research.** Assemble a country-year panel from public sources not previously applied to the disease, compare six model families under three validation schemes, and report the achievable skill including negative results.
2. **Prediction as a service.** Package the trained model as a versioned artefact behind a REST interface, so any consumer can call it without knowing how it works.
3. **Cited explanation.** Build a retrieval-augmented layer that explains a prediction from official guidance, cites every claim, and refuses rather than guesses.
4. **Owner alerts.** Convert a calibrated probability into few enough messages that they are read, using threshold, suppression and deduplication rules.

All four share one requirement: each component must be independently testable and replaceable, and free of any hardcoded reference to glanders.

4. Scope and Limitations

The study is scoped to country-year resolution. District-level modelling for India was envisaged but is not attempted, because the covariates could not be assembled from public sources within the project period. The explanation and alert layers are resolution-agnostic and apply unchanged once a district-level model exists. The model is a population-level tool: it does not diagnose an individual animal and does not replace laboratory confirmation.

Three limitations shape every result. The label comes from published reports, so a negative may mean no disease or no reporting. Occurrence is reconstructed by an extractor not validated against a hand-annotated set. And with 172 positives the uncertainty on every headline metric is substantial, which is quantified rather than hidden.

Chapter 2

Literature Review and Gap Analysis

1. Epidemiology and Surveillance

Glanders is treated as a neglected zoonosis with high case fatality and a real occupational risk to handlers [1, 2]. A recent systematic review and meta-analysis consolidates seroprevalence estimates and identifies management practice and animal movement as the dominant drivers [3]; regional reviews document continued circulation in Brazil [4]. The disease persists where equid populations are large, animals move often and surveillance is weak. The first two are measurable from public data, while the third decides whether the disease is visible at all and is therefore a confounder rather than a covariate.

Indian surveillance is the strongest evidence base. Malik et al. described outbreaks reported between 2006 and 2011 [7], Singha et al. reported serological surveillance for 2015 to 2018 [6], and the national programme covering 2019 to 2023 produced a prevalence estimate with a risk analysis [5]. These establish where glanders occurs and which animal-level factors the risk analysis identified. None builds a predictive model, and none uses population or reanalysis covariates as predictors.

2. Existing Forecasting Systems in India

The closest operational system is NADRES v2, the National Animal Disease Referral Expert System run by ICAR-NIVEDI [39]. It combines outbreak records from AICRP centres with meteorological and remote-sensing covariates and issues district-level forewarnings about two months ahead, using a model family that overlaps with the one used here.

This project differs in four complementary ways. It is glanders-specific, where a national system must spread effort across many diseases. It uses only public data, so anyone can reproduce and audit it. It attaches a cited explanation to each prediction, which matters when the reader is an owner rather than a trained analyst. And it delivers to the owner directly, with rules that suppress routine noise. NADRES has institutional access this project lacks; this project has a level of local explainability that a multi-disease platform does not attempt.

3. Spatial Risk and Machine Learning

Spatial work on glanders is descriptive. Fonseca-Rodríguez et al. applied spatiotemporal cluster detection to Brazilian data [8], and species distribution models have been compared as tools for livestock disease [9]. Cluster detection identifies where cases have already concentrated; it does not forecast which cells will be positive next period, which is the question an early warning system must answer.

Predictive modelling is established for avian influenza. Kane et al. compared ARIMA and random forest models for H5N1 outbreaks [11], Walsh et al. used machine learning to target active surveillance [12], and Yoo et al. and Alshatti et al. built farm-level risk models in Korea and Kuwait [13, 10]. That literature supplies the methodological template used here: assemble occurrence, population and environmental covariates at a common unit, fit tree-based ensembles, and evaluate on held-out periods. The template has been applied to influenza, not to glanders.

The only machine learning work specific to *B. mallei* is spectroscopic identification. Moawad et al. built a Raman assay in which learned models classify spectra [14], and Stöckel et al. reported spectroscopic detection of *B. mallei* and *B. pseudomallei* [15]. Both are laboratory diagnostic problems; neither addresses geographic or temporal distribution, and neither produces a population risk estimate. Using published literature as structured occurrence data has precedent in epidemic intelligence [16, 17], but here event extraction is the primary source of the label rather than a supplementary signal, because official outbreak records are incomplete for many countries. The label therefore inherits the reporting biases of the literature, which is analysed rather than assumed away.

4. Prior Art Audit

To test the novelty claim rather than assert it, a reproducible set of queries was run against Europe PMC. Each row of Table 2.1 is a query count; the query strings and retrieved titles are retained with the project code.

Table 2.1: Europe PMC hit counts for combinations of glanders with each data source and method used in this study.

Query	**Hits**
glanders (baseline corpus)	6674
glanders + machine learning	75
glanders + tree ensembles	40
glanders + forecasting	139
glanders + FAOSTAT	7
glanders + NASA POWER	0
glanders + World Bank indicators	22
glanders + equine population denominator	54
glanders + Europe PMC text mining	378
glanders + chained/stacked models	36
glanders + random forest (risk context)	8
glanders + niche / distribution model	62
glanders + remote sensing / reanalysis	10
glanders + spatio-temporal clustering	463
glanders + gradient boosting	4

The decisive cells pair glanders with NASA POWER, which returns zero records, and with gradient boosting, which returns four, none a glanders risk model.

5. Research and Deployment Gap

Research gap. The literature divides into three disjoint groups. Epidemiological studies describe where glanders occurs; machine learning studies of *B. mallei* classify laboratory spectra; veterinary risk models exist but are built for influenza. No study joins an occurrence corpus, a livestock census, development indicators and reanalysis climatology into a single panel for glanders, and none evaluates a calibrated predictive model on that panel.

Deployment gap. A probability on a dashboard does not reach the person who owns the animals, and a risk score without a stated basis is not something a field officer will act on. The literature stops short of the delivery problem: how a forecast becomes a cited explanation, and how that becomes a message an owner trusts.

The contribution has three parts: a calibrated risk model, evaluated under forward temporal and country-holdout splits and served as a versioned artefact; a retrieval-augmented layer that grounds each prediction in official control guidance, cites every claim and refuses rather than generates; and an alert service whose threshold, suppression and deduplication rules turn a probability into a small number of messages. All three exchange fixed versioned contracts and none hardcodes the disease.

Chapter 3

System Architecture

1. What the Platform Has to Do

The platform answers one question: whether glanders is likely to be documented in an area next period. It produces three outputs from it, a number on a map, an explanation in words and a message to an owner, each read by a different person on a different device, and each able to fail without the others failing. One service serving all three would tie alert delivery to the latency of an LLM call and make the risk model impossible to test alone.

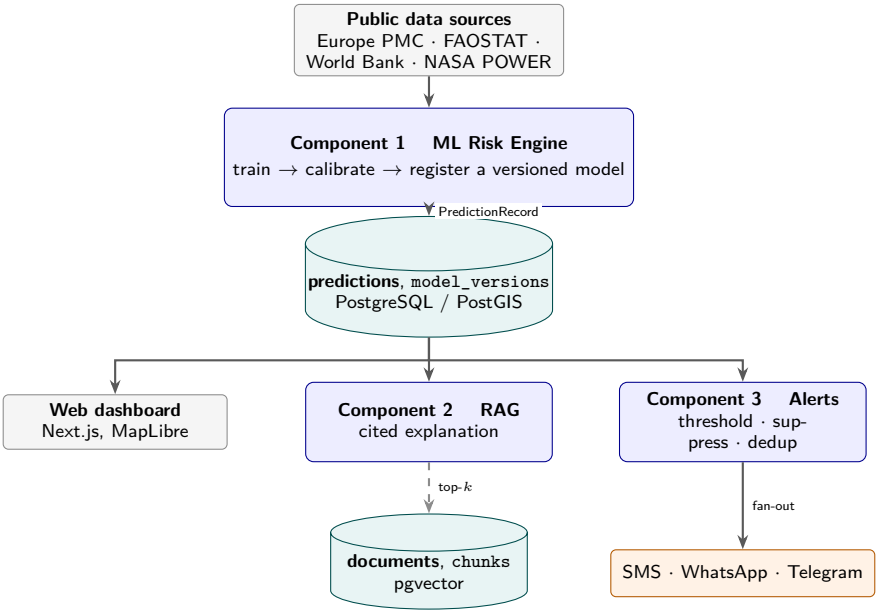


Figure 3.1: The platform decomposed into three components. The ML risk engine writes one `PredictionRecord` per area and period; the dashboard, the RAG layer and the alert service consume it independently. No component reads another component's tables, so any one of them can be replaced without touching the other two.

2. Design Principles

Five rules govern the decomposition, and every later design decision follows from them.

One responsibility per component. The risk engine turns features into a calibrated probability, the RAG layer a prediction into a cited explanation, and the alert service a prediction into a delivered message. None computes what another could.

Contracts before implementations. Components exchange versioned schemas, never shared tables or in-process calls, so each side is built and tested against a fixture. That is what let the RAG layer be developed while the alert service was still being written.

The core is disease-agnostic. A disease is a configuration record: which label extractor, knowledge collection, templates and thresholds to use. A second disease is a configuration change plus a retrain, not a code fork.

Fail soft, never fail silent. A degraded component returns a degraded answer. If the RAG layer is unavailable the alert still goes out without its explanation; if retrieval finds nothing above the relevance floor, the layer returns a refusal plus the model's own SHAP factors. Neither case produces a wrong statement, and neither produces silence.

Every output is traceable. Each prediction carries its model version, each explanation the documents it was grounded in, each alert the prediction and rule outcomes behind it, so a decision can be reconstructed afterwards.

3. The Three Components

Table 3.1: Component responsibilities, interfaces and failure behaviour.

Component	Responsibility	Interface	On failure
ML risk engine	Features to calibrated probability, with attribution	`POST /v1/predict`, `GET /v1/risk-map`	`503` with the champion id; the last good prediction is marked stale.
RAG layer	Prediction plus question to a cited explanation	`POST /v1/explain`, `POST /v1/chat`	Alert text is sent without the explanation; chat returns a refusal.
Alert service	Prediction to a delivered, deduplicated message	`POST /v1/alerts/evaluate`	Alerts queue in Redis and drain on recovery; map is unaffected.

4. Component Contracts

A contract is a JSON schema plus its guarantees, and a narrow contract is what makes a component replaceable. The risk engine emits a `PredictionRecord` with area, period, calibrated probability, risk level, model identifier, a freshness block and the top SHAP factors. Three guarantees follow: the probability is calibrated on held-out data and reads as a frequency; the risk level is derived from it by the threshold configuration, so the two cannot contradict each other; and the factors come from the model, so they survive the RAG layer being unavailable.

The RAG layer takes an `ExplainRequest` and returns an `ExplainResponse` with an answer, resolved citations, a `grounded` flag and a refusal reason; a refusal carries no answer text. The alert service emits an `AlertEvent` with its identifier, area, period, both levels, source prediction, rule outcomes and recipient count. Components share a database but not tables, and spatial columns use PostGIS while embeddings use pgvector.

5. Modularity, Reuse and Status

The decomposition justifies its overhead only if the parts are reusable. A second disease supplies a new gazetteer, guidance collection and thresholds, while the modelling code, which never names glanders, does not change. A second LLM provider is a configuration value, because the RAG layer calls a single-method interface; a second channel is an adapter plus a template. The pipeline is a scikit-learn compatible object [32] exported with a manifest, so any Python process can load it without the API.

The report distinguishes throughout between what has been built and what has been specified: the risk engine is built and evaluated, with every number in Chapter 8 reproducible from the committed scripts, while serving and the RAG and alert layers are specified but not deployed, so no measured performance is reported for them.

Chapter 4

Methodology

1. Data Sources

Four public sources are used: a literature index, a livestock census, a development indicator set and a reanalysis climate archive [38, 35, 37, 36]. None had previously been applied to glanders risk modelling.

Europe PMC supplies the animal history: 6,674 records yielding 263 occurrence events. FAOSTAT supplies animal population, 68,449 records covering seven stock series from 1961 to 2023. The World Bank supplies socio-economic context, ten development indicators and country centroids for 217 countries from 2000 to 2024. NASA POWER supplies environment, six measures of temperature, precipitation, humidity and wind at 211 centroids from 2005 to 2024.

Extracted events rise in volume after 2015, which reflects growth in open access veterinary publishing as much as any change in incidence. This is one reason the label is treated as documented activity throughout.

2. Panel Construction

All four sources are keyed on ISO 3166 alpha-3 code and calendar year, so they join once names are reconciled. FAOSTAT is streamed from its 32 MB archive and filtered to the seven stock series rather than loaded whole. Names are matched through `pycountry` with overrides for forms such as "Türkiye", and World Bank aggregates are excluded by requiring a three-character ISO code, so "World" never enters the panel as a country.

The grid holds 6,700 country-year rows across 268 countries for 2000 to 2024. Three restrictions produce the modelling matrix: only equine-holding countries, defined as a maximum equid stock of at least one thousand; only 2005 to 2023; and only rows with an observable next-year label. The result has 3,078 rows, 44 features and 172 positives.

This yields the modelling matrix, 3,078 rows by 44 features with 172 positives, which is the dataset every model was trained and evaluated on. It is committed as `model_matrix.csv` under `data/processed/`, one row per country-year, with the feature columns in pipeline order and the binary next-year label. The file is written by `src/11_export_model_matrix.py` from the same loader and feature definitions the training script uses, so it cannot drift from what the models saw.

Events are de-duplicated on country, year and source record identifier, so one paper reporting several cases cannot contribute multiple counts. Missing values are median-imputed inside the training pipeline, with a missing-indicator feature per affected column. Missingness is low: 0.00% for FAOSTAT, 1.68% for the World Bank, 2.19% for climate, and 0.99% overall, no column above 5.8%.

3. Label Definition

The target is binary: whether a glanders occurrence was documented in country i during year $t + 1$. Occurrence is extracted from Europe PMC by matching, within one sentence, a country name from a

gazetteer and a term from an occurrence lexicon (*outbreak*, *case*, *detected*, *seropositive*, *surveillance*), recorded for the year in the sentence or the publication year. Because the label depends on publication it is termed *documented activity*: a negative may mean no disease or no report, so the model predicts observable activity, not biological presence.

4. Feature Engineering

Four groups are built from information at year t or earlier. Seven disease-history features cover prior-year occurrence, a five-year total, years since the last event and activity shares. Fourteen population features cover log stocks, equid share, density, growth and species counts. Ten climate features cover temperature, precipitation, humidity and wind, each with an anomaly against the country's own expanding mean, separating unusual conditions from latitude. Eleven socio-geographic features cover income, agriculture, GDP, rural share and connectivity, and largely proxy surveillance capacity.

Three derived features need definition. Equid share is horses, asses and mules over total farm animals. Livestock units weight cattle and camels at 0.8 and small ruminants at 0.1, so one number expresses grazing pressure across species. Years since the last event uses 99 for no event in the window.

Selection ran in two stages. Columns with mutual information below 0.002 and absolute Spearman correlation below 0.02 were removed, then one member of each pair correlated at or above 0.90, plus a column that was exactly the sum of three others, which a tree can construct itself. No projection was applied: with 44 features and 3,078 rows the binding constraint is signal rather than dimensionality.

5. Statistical Association Measures

Four measures screen the features, each from a standard library routine rather than a handwritten implementation, with versions recorded in Section 5. Let $y \in \{0, 1\}$ be the next-year label and n the number of complete pairs.

Spearman's rank correlation (`scipy.stats.spearmanr`) measures monotone association without assuming linearity, which matters because equid stock spans four orders of magnitude:

$$\rho = \frac{\operatorname{cov}\big(R(x), R(y)\big)}{\sigma_{R(x)}\,\sigma_{R(y)}}.$$

Mutual information (`sklearn.feature_selection.mutual_info_classif`, $k = 5$) captures non-monotone dependence that a rank correlation misses, estimated by the Kraskov–Stügbauer–Grassberger k-nearest-neighbour form:

$$I(X;Y) = \iint p(x,y) \log \frac{p(x,y)}{p(x)\,p(y)}\,\mathrm{d}x\,\mathrm{d}y, \qquad \hat{I} = \psi(n) - \big\langle \psi(n_x{+}1) + \psi(n_y{+}1) \big\rangle + \psi(k),$$

with ψ the digamma function and n_x, n_y per-point neighbour counts.

Cramér's V (`scipy.stats.chi2_contingency`) measures association between a categorical feature and the binary label, placing income band and region on one scale:

$$V = \sqrt{\frac{\chi^2}{n\,\min(r-1,\,c-1)}}.$$

Variance inflation (`numpy.linalg.lstsq`) quantifies how far one feature can be reconstructed from the others, and so how unstable its coefficient would be:

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 comes from regressing x_j on the remaining features. This is the measure behind the collinearity result in Chapter 8.

6. Models

Six families are compared. Logistic regression is the additive baseline with balanced class weights. A single shallow decision tree is included because a non-specialist can read it. Random forest and extra trees are bagged ensembles [20], differing in split randomisation; XGBoost and LightGBM are boosted ensembles [18, 19]. Tuning used grouped cross-validation on country. Neural networks were excluded: with 172 positives and 44 features a network has more parameters than examples and would memorise the training period.

7. Evaluation Design

Three schemes answer different questions. A temporal holdout trains on 2005 to 2017 and tests on 2018 to 2023, with the test period never observed during fitting, which answers whether the model can be used in following years. Leave-country-out uses five folds grouped by country, so every prediction is for an unseen country; it answers whether the model transfers and it governs selection, being the harsher test and the closer approximation of deployment. Repeated stratified cross-validation, five folds three times, is a stability check.

Precision-recall AUC is the primary metric: at 5.59% prevalence ROC-AUC is dominated by the true-negative mass and overstates performance [22, 23], so it is reported only for comparability. Brier score and expected calibration error are reported because an uncalibrated probability cannot support a threshold-based alert [25, 26]. Precision, recall, specificity, F1 and the Matthews correlation use a threshold tuned on training data only, with bootstrap percentile intervals over one thousand resamples. Grouped rather than random splitting is essential, since a random row split would let the model memorise country identity [24].

8. Leakage Controls

Six controls apply: features use year t or earlier and only the target uses $t + 1$; the split is chronological; hyperparameters are tuned inside training folds; calibration is fitted on training folds only; the alert threshold is fixed on training; and imputation values and missing indicators are generated inside the pipeline.

Chapter 5

MLOps and Model Serving

1. From a Notebook to a Service

A trained model is not a component until it can be called, versioned, monitored and replaced. Three things must be true first. The model must be a portable artefact, so the object scored in evaluation is the object served, with nothing re-fitted in between. Every prediction must name the version that produced it. And the service must degrade predictably, so a bad release or a missing feature does not produce confident nonsense. The project is a sequence of numbered scripts, each producing artefacts consumed by the next, with every intermediate output cached so any stage can be re-run alone. That is also what makes the pipeline inspectable: a reader can run one stage and check it before trusting the next.

2. The Model Bundle

The unit of deployment is a directory, not a pickled object, because a pickle alone cannot say what its inputs mean.

```
gw-xgb-2026.11.1/
  model.joblib        # fitted pipeline, predict_proba
  manifest.json       # feature order, dtypes, calibration, metrics
  thresholds.json     # level boundaries and hysteresis margin
  SHA256SUMS
```

The manifest is what makes the bundle reusable. It records the ordered feature list with dtypes, so serving rejects a request with the wrong columns before any inference happens. It records the calibration method and achieved error, so a consumer knows whether to read the output as a probability or only as a rank. It records the evaluation metrics, so the number in the report and the number in the registry are the same. Because the bundle is a directory with a manifest, the API, a batch job and a notebook load it with identical behaviour.

Every training run also writes a record with the git commit, data version, feature list and hash, hyperparameters and the full metric set for all three schemes. Promotion is decided on leave-country-out precision-recall AUC rather than on the more flattering temporal holdout. A candidate becomes champion only if it beats the incumbent by more than the bootstrap interval width, its expected calibration error is below the deployment ceiling of 0.05, its feature schema matches what serving can supply when checked by name and order, and it has scored a shadow period without a prediction outside the plausible range. The previous champion is retained, so promotion is reversible, and a bundle failing its checksum or schema check is replaced by the previous champion rather than served.

3. Serving Design

The interface is deliberately coarse: a caller asks for a prediction for an area and period and receives the full record, rather than assembling one from several calls.

Table 5.1: HTTP interface of the risk engine.

Endpoint	Method	Purpose and contract
`/v1/predict`	POST	Score one area and period from a supplied or stored feature vector. Returns one `PredictionRecord`.
`/v1/predict/batch`	POST	Score up to 500 area-period pairs, with per-item errors rather than failing the request.
`/v1/risk-map`	GET	GeoJSON of the latest predictions for a period, filtered by risk level. Serves the dashboard directly.
`/v1/models`	GET	Registry listing: model id, promotion date, metrics, feature-schema hash.
`/v1/models/{id}`	GET	The manifest of one version, for inspection by a consumer.
`/v1/health`	GET	Liveness and readiness, where readiness checks that a champion bundle is loaded and its schema resolves.

Every response carries the `model_id` that produced it and a `data_freshness` block. The alert service reads that block before sending anything, so a prediction computed on stale upstream data cannot produce a message.

The order of operations is a reliability choice. Validation runs first, so a malformed request costs a schema check rather than a model load. The cache is consulted before inference because risk maps are read far more often than recomputed. Inference is deterministic for a fixed bundle, so a cache key of model id, area and period is safe, and the model id is part of the key rather than a version counter so promotion invalidates the cache. Responses are written back to the predictions table as well as cached, so the map does not depend on a warm cache.

The endpoint is specified against a budget rather than measured, since the service is not deployed: 40 ms on a cache miss and 4 ms on a hit, of which feature retrieval takes 25 ms and inference 8 ms. The explanation endpoint has a separate budget of two seconds because it includes generation, which is why the alert service never calls it inline. A message must not wait on an LLM.

4. Monitoring and Retraining

Three signals are monitored. Service health covers request rate, error rate and latency percentiles per endpoint on Prometheus and Grafana, catching the least interesting failure. Data drift compares each feature's current distribution with the training one using a population stability index and a Kolmogorov-Smirnov statistic, since a tree extrapolating outside its training range returns a confident answer with no basis. Calibration decay tracks the mean predicted probability against the observed event rate: a model that keeps saying 0.2 while the observed rate falls to 0.05 has lost calibration even with its ranking intact.

Retraining follows from the third signal. Recalibrate when the observed-to-predicted ratio leaves a band for two consecutive periods, and retrain when a new year of labels is available and the champion's leave-country-out score has degraded beyond the bootstrap interval.

The pipeline runs linting, unit tests and a smoke test on every push. Unit tests cover what can be checked without data: schema validation, the level-threshold function including hysteresis at the boundary, idempotency key derivation, and the citation parser. A further test loads the committed bundle, scores a fixed fixture and asserts the output to four decimals, which catches the most common silent failure here, where a dependency upgrade changes numerical behaviour and every downstream number moves unnoticed.

The full computation runs in about 4.5 minutes on an Apple M4 processor with 10 cores and 16 GB of unified memory without a graphics accelerator. Per-family fitting time at the country-year level ranges from 1.9 seconds for the decision tree to 50.7 seconds for LightGBM. One module-level seed governs fitting, cross-validation, bootstrap resampling and SHAP sampling, with software versions Python 3.12.13, scikit-learn 1.9.1, XGBoost 3.4.1, LightGBM 4.7.0, SHAP 0.54.1, NumPy 2.5.4, pandas 3.1.1, SciPy 1.18.1, and the bundle smoke test runs single-threaded because tree ensembles are not bitwise reproducible across thread counts.

Chapter 6

The RAG Explanation Layer

1. The Problem This Layer Solves

A calibrated probability of 0.24 is actionable for a veterinary officer and meaningless to an equine owner. The owner needs to know what happened and what to do next; the officer needs the factors behind the number, because a forecast they cannot interrogate is one they will not act on.

Asking a large language model is the obvious answer and the obvious hazard. A model asked about glanders answers from training data that may be outdated, may conflate glanders with melioidosis, and may invent a treatment protocol; for a zoonosis under statutory control, an invented instruction is worse than none. Retrieval-augmented generation [27] changes what the model may use: it is given a small set of passages from official guidance plus the prediction's own numbers, and required to answer only from those, so each claim traces to a passage the reader can see.

The goal is narrow and testable: an answer a veterinary officer would accept as consistent with the National Action Plan, and a refusal rather than a guess when the corpus does not support one.

Fine-tuning would produce fluent text with no citations, no way to separate guidance from the base model's priors, and knowledge frozen at training time. Retrieval keeps knowledge in a corpus a re-ingestion can update, makes every answer auditable, and allows a refusal path, at the cost of latency and possible retrieval failure.

2. Knowledge Base

The corpus is restricted to authoritative sources, because the layer's value is that it cites them. Each document carries an authority tier, used as a tie-breaker during retrieval.

Table 6.1: Knowledge base, by authority tier.

Source	Tier	Use
DAHD National Action Plan for Glanders, 2025 [34]	1	Case definitions, surveillance obligations, movement control, quarantine, notification duties.
WOAH and WHO disease information sheets	1	Disease biology, transmission, zoonotic risk, official status.
ICAR and national laboratory guidance	2	Diagnostic sampling, laboratory referral.
Open-access research papers	3	Epidemiology, risk factors, regional situation.

Tier 3 may inform an explanation but never overrides tier 1 on a control instruction. Conflicts resolve to the higher tier and are logged, because a silent contradiction between sources is the failure this layer exists to prevent.

3. Ingestion, Embedding and Retrieval

Ingestion runs offline as a job and stores each chunk with enough metadata to be cited and filtered. Chunking is section-aware rather than fixed-width, because a fixed window cuts control instructions in half and lets the model complete a truncated sentence. The splitter respects headings first, then paragraphs, falling back to a sliding window only past the ceiling. Chunks target 400 to 700 tokens with 15 per cent overlap inside a section, and sections under 80 tokens merge into the parent. The full section path is retained, for example `NAP-2025 > 5.2 Movement control`, so a citation points to a subsection rather than a 90-page document.

Chunks are embedded with a compact sentence-transformer [28] producing 384-dimensional vectors, with a multilingual variant for Hindi because an English-only embedding degrades on Devanagari. Vectors live in pgvector beside the relational tables, so re-ingestion and prediction share one transaction boundary. The index is HNSW with cosine distance [29]; the reranker corrects recall at the head of the list, and the index keeps query cost flat as the corpus grows.

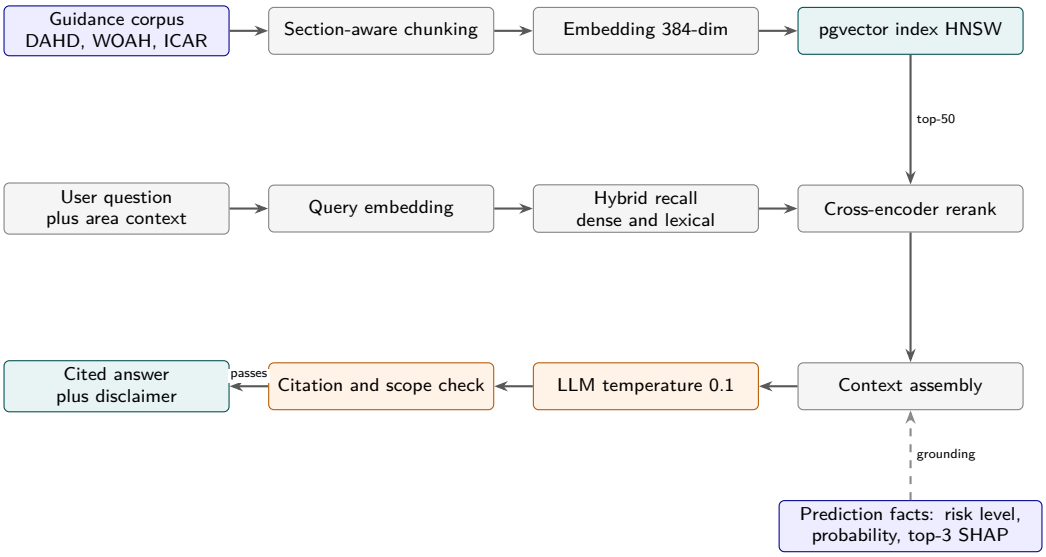


Figure 6.1: The RAG layer. Ingestion (top row) is an offline job; retrieval, assembly and generation (below) run per request. Predicted risk and SHAP factors are injected as structured facts, so the answer explains this forecast rather than glanders in general. The citation check sits after generation and can return a refusal instead of an unsupported answer.

Retrieval is hybrid, because dense and lexical search [30] fail differently here. A dense query for "movement control" may miss a passage using "restriction of equid movement", where a lexical match succeeds; conversely a lexical query for a rare acronym retrieves noise a dense query ignores. Both lists are fused by reciprocal rank fusion [31], which combines rankings without requiring comparable scores:

$$\mathrm{RRF}(d) = \sum_{i \in \{\text{dense},\text{lexical}\}} \frac{1}{k + \mathrm{rank}_i(d)}, \qquad k = 60$$

The fused top fifty are reranked by a cross-encoder scoring each passage against the question jointly. It is far slower per pair than an embedding dot product, which is why it runs on fifty candidates rather than the corpus, and the top five become the context.

Table 6.2: Retrieval parameters.

Stage	Setting	Reason
Dense recall	top 30, cosine	Semantic paraphrase coverage
Lexical recall	top 30, BM25	Exact terms, acronyms, section titles
Fusion	reciprocal rank, $k = 60$	Scale-free combination of two rankings
Rerank	cross-encoder, top 50 to 5	Precision at the head of the list
Relevance floor	rerank score below 0.30 discarded	Prevents padding the context with weak passages

The relevance floor drives the refusal path: if fewer than two passages clear it, the layer returns a refusal with the SHAP factors, because marginal passages produce a confident answer built on nothing.

4. Context, Generation and Guardrails

The prompt has three parts in fixed order: a system instruction, the retrieved passages with identifiers, and the structured prediction facts. Those facts make the answer specific to this forecast rather than to glanders in general, and are injected as structured data so the model cannot restate an association as a cause. The instruction states four constraints: answer only from the passages, cite every claim, never diagnose an individual animal, and state that the output is a population-level forecast.

Generation uses temperature 0.1 and a bounded output length, chosen for consistency rather than caution: two officers asking the same question about the same area should receive the same instruction. The provider sits behind a single-method interface, so one retrieval, grounding and citation path serves a local Ollama model or a hosted API, and switching model cannot silently disable the guardrails.

The layer is designed to refuse, and four checks can terminate a response.

Table 6.3: Guardrails and their failure behaviour.

Check	What it verifies	On failure
Retrieval floor	At least two passages above the relevance threshold	Refusal with the top SHAP factors, which need no retrieval
Citation resolution	Every cited identifier exists in the retrieved set	Answer rejected and regenerated once, then refusal
Diagnosis scope	No individual-animal diagnosis or treatment prescription	Response rewritten without the clinical claim, and the attempt logged
Out-of-scope	Question is about glanders control, not unrelated topics	Refusal stating what the layer covers

The citation check is mechanical: the answer is parsed for identifiers and each is looked up in the retrieved set, and one that does not resolve means the model cited a document it was not given, which is a hallucination whether or not the sentence is true. Asking a model to be accurate is not a control; checking its output is.

5. Evaluation

Five quantities are defined below, following the project proposal. None is reported as a measurement, because the layer has not been run against the corpus; measuring them is future work (Chapter 9).

Retrieval quality. Recall@k, the fraction of hand-built questions whose known relevant passage appears in the top k results, at $k = 5$, with mean reciprocal rank, which rewards returning it near the top rather than merely inside the cut-off. This isolates retrieval from generation, since no generation quality helps if the right passage is never retrieved.

Faithfulness. The fraction of claims the retrieved context supports, scored by a judge model and reviewed by hand on a sample; the RAGAS-style metric that catches a model citing a passage while saying something the passage does not say.

Answer relevance. Whether the answer addresses the question asked rather than reciting correct but tangential guidance, scored on the same sample, since an answer can be grounded and still unhelpful.

Citation precision. The fraction of citations resolving to a passage that supports the sentence, distinct from resolution, which only checks existence.

Refusal behaviour. The refusal rate on out-of-scope questions and the false refusal rate in scope: a layer that never refuses is unsafe and one that always refuses is useless.

Deployment targets, so the claim is falsifiable: recall@5 above 0.85, faithfulness and answer relevance above 0.90 and 0.85, citation precision above 0.95, a false refusal rate below 10 per cent, and p95 latency under two seconds.

Chapter 7

The Alert Service

1. Why the Alert Is the Hard Part

The risk engine produces a probability and the RAG layer explains it, but neither reaches the person who owns the animals, whose only interaction is a message on a phone competing with everything else on it. The problem is not how to send a message but how to send few enough that the ones which arrive are read; a system that warns about every area every month fails silently at the moment it was built for.

The service is event-driven: the prediction job publishes `predictions_ready` and the evaluator consumes it, so an alert cannot be sent for a period whose predictions were not recomputed. For each subscribed area the evaluator reads the new prediction and the previous alert state, and those two values are the entire input to the rules. It reads no features or model internals, so a model change can affect alert behaviour only through the probability it produces.

2. Rule 1: Threshold with Hysteresis

The calibrated probability maps to three levels. The boundaries are configuration values derived from the operating curve in Chapter 8, not defaults and not 0.5.

Table 7.1: Level thresholds and their basis.

Level	Band	Meaning for the recipient	Basis
LOW	below the medium boundary	No message. Routine monitoring.	Below the budget where precision collapses
MEDIUM	medium to high boundary	Prepare; check animals; note the veterinary contact	Operating point where recall exceeds 80 per cent
HIGH	above the high boundary	Act: restrict movement, isolate suspects, inform the authority	Top of the operating curve, precision above 0.5

Hysteresis addresses a specific defect. With a single boundary, an area whose probability oscillates around the threshold produces an alert every period, because each crossing is a state change. Two boundaries are therefore used: an area enters HIGH at the high threshold and leaves only when the probability falls below that threshold minus a margin.

$$\text{enter HIGH at } p \geq T_H, \qquad \text{leave HIGH only at } p < T_H - \delta, \qquad \delta = 0.05$$

The margin is configurable per disease. A larger margin gives a more stable state at the cost of slower de-escalation, which suits a disease where a missed warning costs more than a late all-clear.

3. Rule 2: Suppression

Suppression decides whether a message is warranted at all, given that the level has changed. Six conditions apply and any one can stop the message.

Table 7.2: Suppression conditions.

Condition	Rule	Rationale
State change	Send only on a transition, not while the level is held	A monthly reminder at the same level carries no new information
Cooldown	No repeat of the same level within a configurable window	Prevents oscillation at a boundary from producing repeated messages
Escalation bypass	An upward transition ignores the cooldown	A worsening must never be delayed by a timer
Per-user daily cap	At most one message per recipient per day	Protects against several areas escalating at once
Quiet hours	Nothing sent between 21:00 and 07:00 local; queued instead	A warning that wakes a household at night is resented, and is not actionable then
Stale-data guard	No send if `data_freshness.passes` is false	An alert about a prediction computed on stale inputs is worse than no alert

A manual mute per area lets an officer silence a zone under active surveillance without disabling the service. De-escalation is separate: when a level falls, an optional short all-clear is sent, since a recipient told to act needs to be told when to stop. Sending it is a configuration choice, because an all-clear costs a message but prevents a restriction being maintained indefinitely.

4. Rule 3: Deduplication

The service must be safe to re-run: jobs are retried, deployments restart workers, and a model may be re-scored for the same period, and none may produce a second message. Deduplication is enforced at three levels, each with its own key and database constraint.

Table 7.3: Deduplication levels.

Level	Key	What it prevents
Alert	`(area_id, period, level)`	A re-run of the job creating a second alert for the same period
Recipient	`(alert_id, subscription_id)`	One subscriber receiving the same alert through two fan-out passes
Send	`(alert_id, subscription_id, attempt)`	A retry after a timeout delivering a message that had in fact succeeded

The first two are unique database constraints, so protection does not depend on application logic being correct. The alert identifier is derived deterministically from its key rather than at random, so two concurrent workers computing the same alert arrive at the same identifier and one loses the insert. The send-level check reads the delivery log before each attempt, handling the case where a provider accepts a message, the response times out, and the retry would deliver it twice.

5. Fan-Out, Channels and Delivery

Once an alert record exists, one delivery job per subscription is queued through Redis. A channel adapter receives a rendered message and a recipient and returns a delivery status, with adapters for SMS, WhatsApp and Telegram, the last used for demonstration because it needs no commercial onboarding. Retries use exponential backoff with jitter: a permanent failure such as an invalid number is not retried, a transient one such as a provider timeout is, and the distinction is logged, because a silent permanent

failure leaves a subscriber who believes they are covered and is not. Templates are separate from the rules engine, so adding a language adds a template set rather than touching the rules.

The same alert through a richer channel carries the explanation paragraph from the RAG layer with its citations. The rule that decides whether to send is identical; only the adapter and template differ, which is what lets one alert serve a feature phone and a smartphone without duplicating the decision logic.

6. Evaluation of the Alert Layer

The service is evaluated as a warning system, not a messaging system. The proposal states the criterion as missed events against false alerts, and the four quantities below make that measurable. The first two come from historical predictions and labels; the last two need a deployed service, so no measured value is reported.

Alert precision and false-alert rate. Precision is the fraction of HIGH area-periods followed by a documented event. Its complement, the false-alert rate, is the share of messages sent where nothing was reported, and it drives alert fatigue: at a fixed threshold the two are one number seen from opposite sides.

Event coverage and missed-event rate. Coverage is the fraction of observed events preceded by an alert at a given budget, which is the operating curve in Chapter 8; its complement is the missed-event rate. The two trade against each other, and the thresholds are read from that curve.

Noise per recipient. Messages per subscriber per year, which predicts whether the service will still be read in its second year.

Delivery reliability. The fraction of queued deliveries reaching terminal success within the retry window, per channel, since channels fail differently.

Chapter 8

Results and Discussion

1. The Risk Model

1.1 Dataset Characteristics

The modelling matrix holds 3,078 country-year observations, of which 172 are positive: a prevalence of 5.59% and an imbalance of 16.9 to one. 56 of 162 countries record at least one positive, 30 of them in two or more years, and the top five hold 35.5% of all positives.

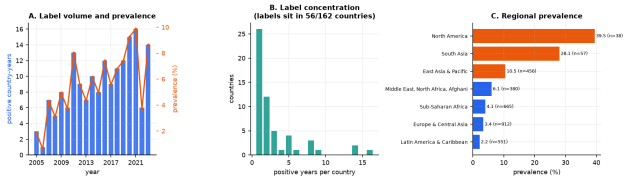


Figure 8.1: Label anatomy. Panel A gives annual positive counts against prevalence on the secondary axis. Panel B gives the distribution of positive years per country. Panel C gives prevalence by World Bank region.

Three properties shape the design. The imbalance, visible as the gap between the count and prevalence axes in panel A, rules out accuracy as a metric. The persistence in panel B, with 30 of 56 positive countries recurring, justifies historical features. The regional concentration in panel C is the first indication that the label tracks reporting capacity, since those countries operate reference laboratories.

1.2 Model Comparison

Table 8.1 reports discrimination under all three schemes, and Figure 8.2 plots the same comparison.

Table 8.1: Discrimination at the country-year level. Brackets give 95 per cent bootstrap intervals on the temporal holdout; Brier score and expected calibration error are reported in Section 8.1.3.

Model	PR-AUC (time)	95% CI	ROC-AUC (time)	PR-AUC (country)
Logistic regression	0.454	[0.34, 0.58]	0.812	0.223
Decision tree	0.389	[0.29, 0.50]	0.861	0.267
Random forest	0.554	[0.43, 0.66]	0.870	0.304
Extra trees	0.530	[0.42, 0.64]	0.854	0.363
XGBoost	0.485	[0.37, 0.60]	0.844	0.281
LightGBM	0.496	[0.38, 0.61]	0.841	0.339

Against a base rate of 5.59%, the strongest temporal result of 0.554 is an approximately 10-fold lift. Every tree-based model beats the logistic baseline and the single tree is weakest, so the relationships are interactive. The more important pattern is the decline under leave-country-out validation: for Random Forest the score falls from 0.554 to 0.304, beyond the bootstrap interval width for every family. That is the signature of learning country-specific structure, and the figure to quote for deployment.

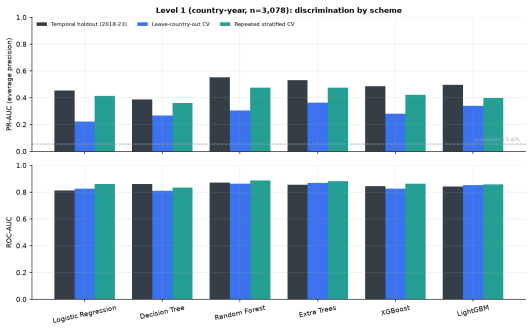


Figure 8.2: Discrimination by model and validation scheme. The gap between repeated cross-validation and leave-country-out validation is the cost of generalising to an unseen country.

1.3 Calibration

Predictions are well calibrated, with an expected calibration error of 0.016 and a Brier score of 0.0535: a stated risk of twenty-four per cent corresponds to an observed frequency near twenty-four per cent, which is what makes a threshold-based alert defensible.

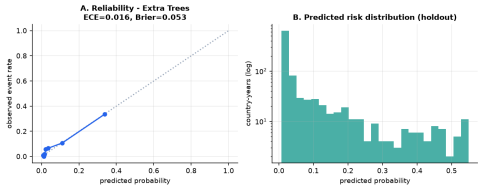


Figure 8.3: Reliability diagram for the selected model on the temporal holdout, with the distribution of predicted risk. Points on the diagonal are perfectly calibrated.

1.4 Subgroup Analysis

Table 8.2 reports leave-country-out performance by region. Skill is lowest in Sub-Saharan Africa and in low-income countries, where early warning would help most; the cause is label sparsity, since few events are recorded in those strata.

Table 8.2: Leave-country-out performance by World Bank region.

Region	Observations	Events	ROC-AUC	PR-AUC
East Asia & Pacific	456	48	0.865	0.396
Europe & Central Asia	912	31	0.826	0.163
Latin America & Caribbean	551	12	0.903	0.381
MENA, Afghanistan and Pakistan	380	23	0.834	0.220
Sub-Saharan Africa	665	27	0.699	0.073

2. The Alert Service as a Warning System

The alert layer turns the calibrated probability into messages, so its economics are reported separately from model skill.

Alerting the six per cent of country-years at highest predicted risk captures about half of all observed events, and twenty-eight per cent captures eighty per cent (Figure 8.4). Uniform alerting would need full coverage for the same recall, so the model is roughly eight times more efficient than arbitrary selection at the fifty per cent point.

For the retrieval-augmented layer no measured results are reported: it is specified in Chapter 6 but has not been run against the corpus, so its five metrics have no values.

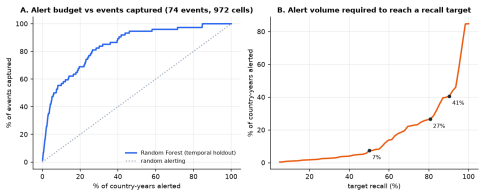


Figure 8.4: Alert economics. Panel A gives the fraction of events captured as the alert budget grows, against random alerting. Panel B gives the alert volume required to reach a target recall.

3. Discussion

The risk model has predictive skill: a precision-recall AUC of 0.554 against a base rate of 5.59% is not explainable by chance, survives a forward-looking split, and is calibrated to within two points. It does not transfer intact: leave-country-out performance is roughly forty per cent lower, so a new country needs local validation first. The signal it learns is surveillance rather than biology, since the highest features are population and connectivity measures, the label concentrates in countries with reference laboratories, and the best-scoring cells are India, the United States and Australia.

Chapter 9

Conclusion and Future Work

1. Conclusion

This project set out to build a usable warning system for glanders: to establish whether population-level risk could be predicted from public data, and how a prediction reaches the person who owns the animals. Four sources new to the disease formed a country-year panel across 162 equine-holding countries, and six model families were trained under three validation schemes. Around the model, delivery was decomposed into three components with fixed contracts: a risk engine, an explanation layer and an alert service.

Prediction is possible, and the performance is moderate: on a forward-looking temporal holdout the strongest model reaches a precision-recall AUC of 0.554 against a base rate of 5.59%, an approximately 10-fold lift with a calibration error of 0.016, capturing half of all events while alerting six per cent of country-years. The skill is fragile across geography, though: leave-country-out performance is about forty per cent lower, and the decline concentrates where early warning would help most.

What the model learns is less encouraging. Three analyses converge: the highest-ranked features are population size and reporting capacity rather than biological drivers, the label concentrates in countries operating reference laboratories, and the best-scoring cells have the strongest publishing record. It predicts where glanders will be documented, not where it is.

2. Future Work

Reported activity. The most important extension is to model surveillance capacity as a quantity, as an offset on publication volume or a joint detection-and-occurrence model; otherwise gains in score reflect reporting, not disease.

The label extractor. The 263 extracted events have never been checked against a hand-annotated set; annotating a stratified sample would give the extractor's own precision and recall.

Resolution and drift. District-month resolution for India matches the scale at which control is applied, but public district covariates would need assembling. The calibration here is one point estimate, so production also needs drift monitoring and a retraining trigger.

Measuring the explanation layer. Its retrieval, guardrail and protocol targets are specified but unmeasured. Running it against the corpus would give recall@5 and mean reciprocal rank for retrieval, faithfulness and citation precision for grounding, and the out-of-scope refusal rate against the false refusal rate in scope. Two metrics belong on top: answer usefulness, rated by district officers on real predictions rather than a hand-built set, and citation age, which shows whether answers lean on guidance since revised.

Measuring the alert service in the field. Four quantities decide whether it survives a second year: messages per subscriber per year for maintainability, delivery reliability per channel for coverage, the fraction of HIGH alerts followed by a documented event for trust, and the time from alert to owner contact for outcome. None can be measured without subscribers.

A WOAH WAHIS record set [33] would validate the literature-derived label where the two overlap, and a cost-sensitive threshold would match the operating point to programme priorities.

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